

Ex.no: 9	MINI PROJECT - II
	SOCIAL MEDIA ANALYTICS

Aim:

To design, store, analyze, and visualize social media data using **SQL Server, Power BI, and Tableau** to derive meaningful insights into views, likes, comments, shares, engagement, content performance, platform performance, audience segments, creator performance, and social media trends.

Algorithm:

1. Create a financial analytics database in SQL Server.
2. Create a Social Media Analytics database in SQL Server.
3. Create relational tables for Platforms, Creators, Content, and Posts with appropriate primary and foreign keys.
4. Populate the database with platform, creator, content, and social media post information.
5. Prepare post-level data containing post dates, platforms, creators, content types, views, likes, comments, and shares.
6. Connect the SQL Server database to Power BI for data modeling and analysis.
7. Validate the data types and prepare the required fields for analysis.
8. Create DAX measures for Total Views, Total Likes, Total Comments, Total Shares, Total Posts, Total Engagement, and Engagement Rate.
9. Develop interactive Power BI visualizations to analyze engagement trends, platform performance, content performance, regional performance, audience segments, and creator performance.
10. Connect the same database to Tableau and establish relationships between the tables.
11. Create Tableau calculated fields for views, likes, comments, shares, total engagement, and engagement rate.
12. Develop Tableau worksheets and an interactive dashboard using KPIs, charts, and filters.
13. Apply filters for Year, Platform, and Content Type to enable interactive analysis.
14. Analyze the dashboards to identify meaningful social media performance patterns.
15. End the process.

Architecture:



Program:

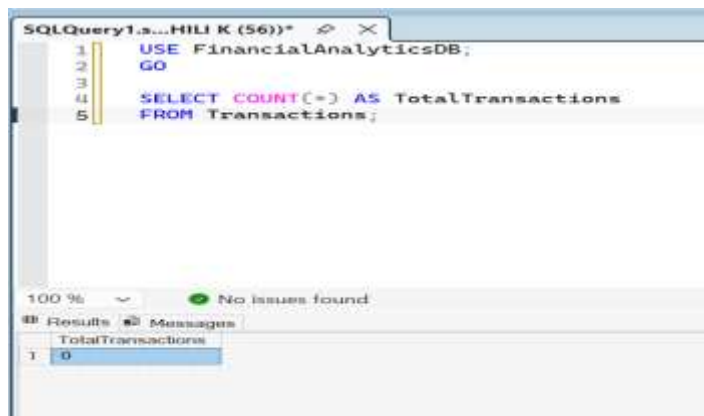
SQL Server – Database Creation

```
CREATE DATABASE SocialMediaAnalyticsDB;
```

```
GO
```

```
USE SocialMediaAnalyticsDB;
```

```
GO
```

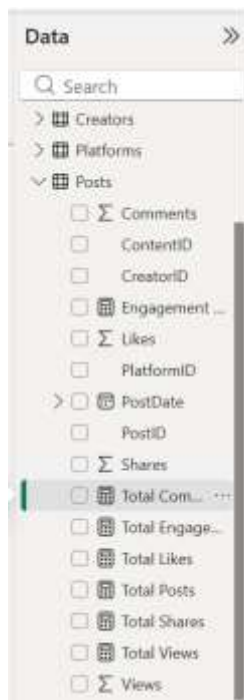


1- Power BI – Data Visualisation

The **SocialMediaAnalyticsDB** database was connected to Power BI Desktop using the SQL Server connector.

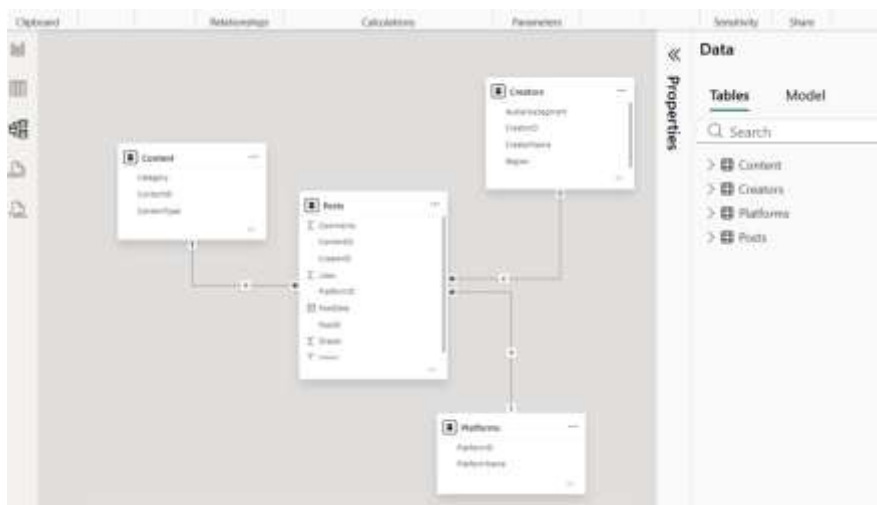
The following tables were imported:

- Platforms
- Creators
- Content
- Posts



Power BI – Data Model

A relational analytical model was created in Power BI.



Power BI – DAX Measures

Total Views

Total Views = SUM(Posts[Views])

Total Likes

Total Likes =SUM(Posts[Likes])

Total Comments

Total Comments = SUM(Posts[Comments])

Total Shares

Total Shares = SUM(Posts[Shares])

Total Posts

Total Posts = COUNTROWS(Posts)

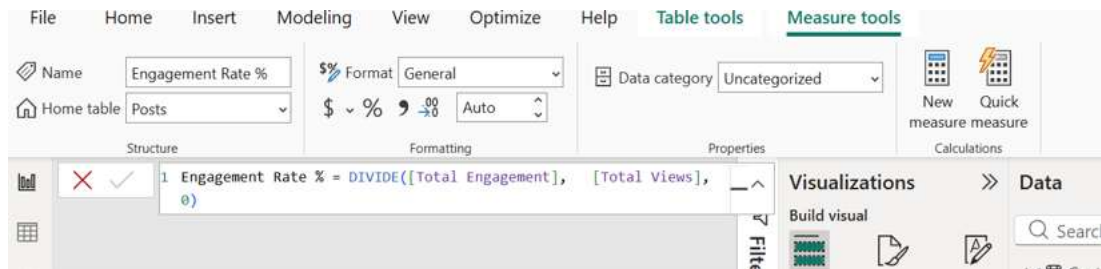
Total Engagement

Total Engagement = [Total Likes] +[Total Comments] +[Total Shares]

Engagement Rate

Engagement Rate % =

DIVIDE(
[Total Engagement],
[Total Views], 0
)

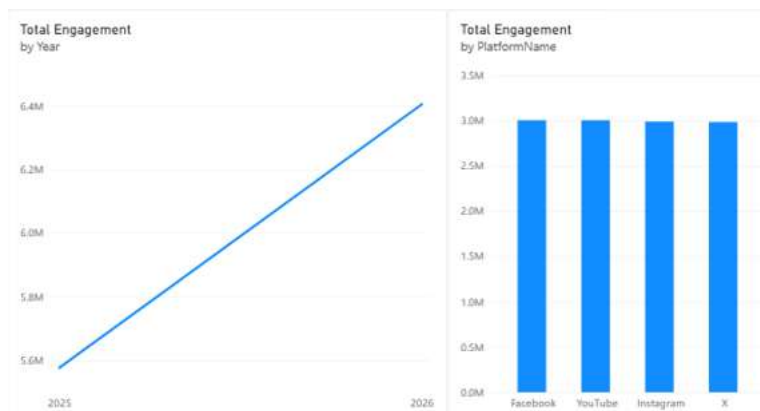


Data Visualization Using Power BI

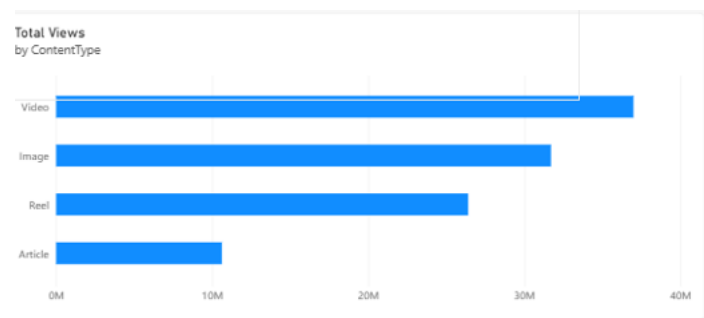
KPI Cards: Card



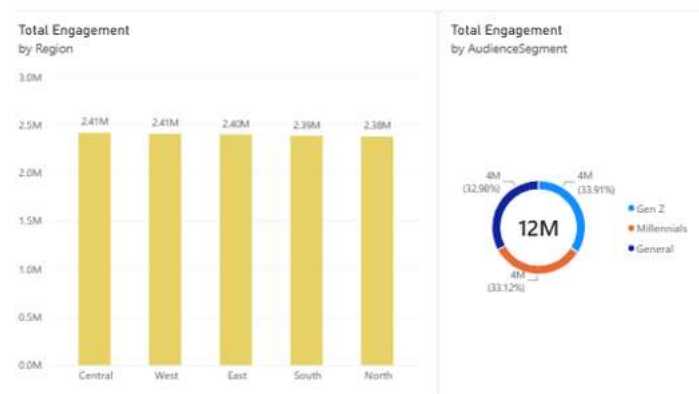
Engagement Trend: Line Chart , Engagement by Platform: Column Chart



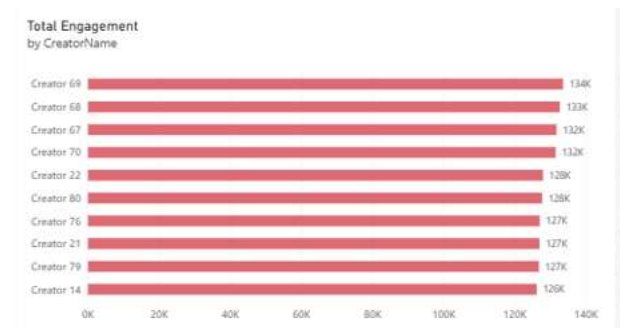
Views by Content Type: Bar Chart



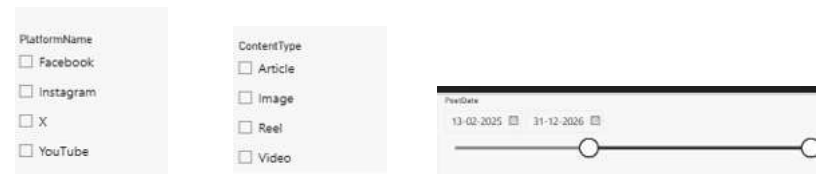
Engagement by Region: Column Chart , Engagement by Audience Segment: Donut Chart



Top 10 Creators: Bar Chart



Interactive Slicers



Power BI Dashboard



2- Tableau – Data visualization

The same SocialMediaAnalyticsDB database was connected to Tableau using the SQL Server connector.

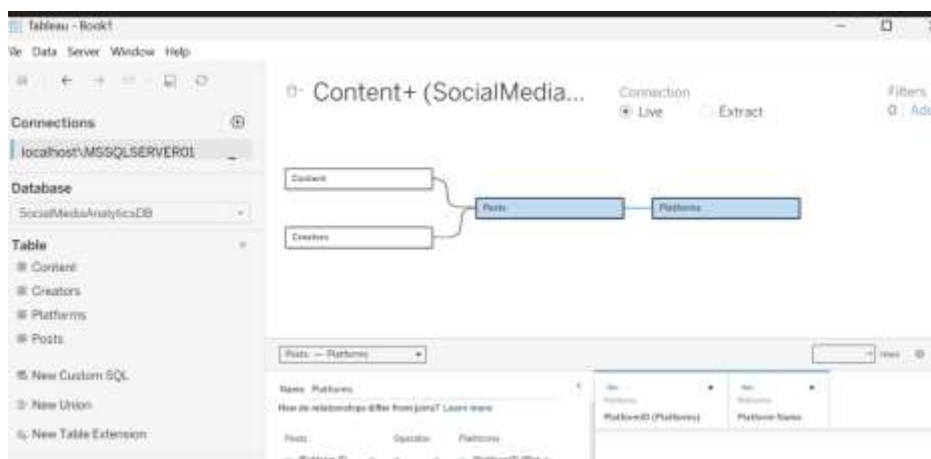


Tableau – Calculated Fields

Total Views

SUM([Views])

Total Likes

SUM([Likes])

Total Comments

SUM([Comments])

Total Shares

SUM([Shares])

Total Engagement

SUM([Likes]) + SUM([Comments]) + SUM([Shares])

Engagement Rate %

(SUM([Likes]) + SUM([Comments]) + SUM([Shares])) / SUM([Views])

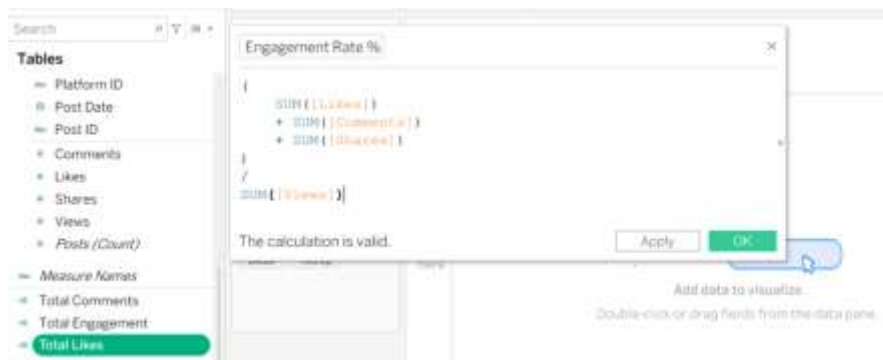
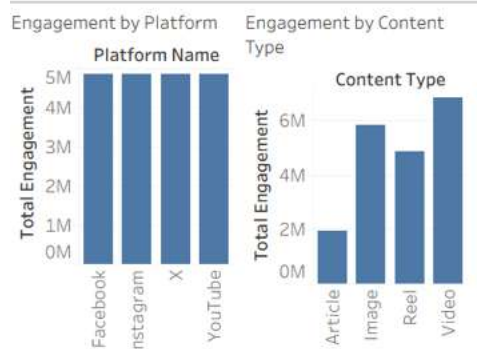
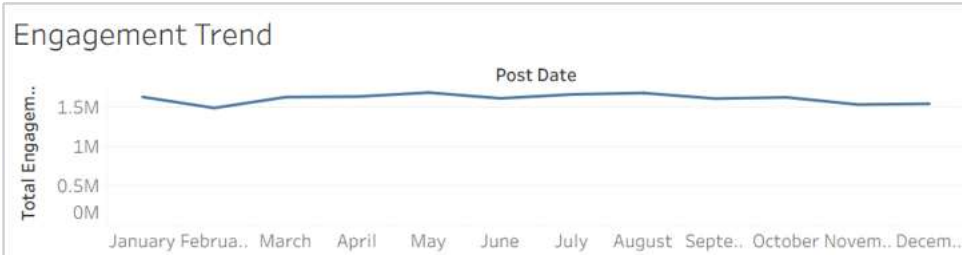


Tableau Dashboard

The Tableau worksheets were combined into an interactive Social Media Performance Dashboard.

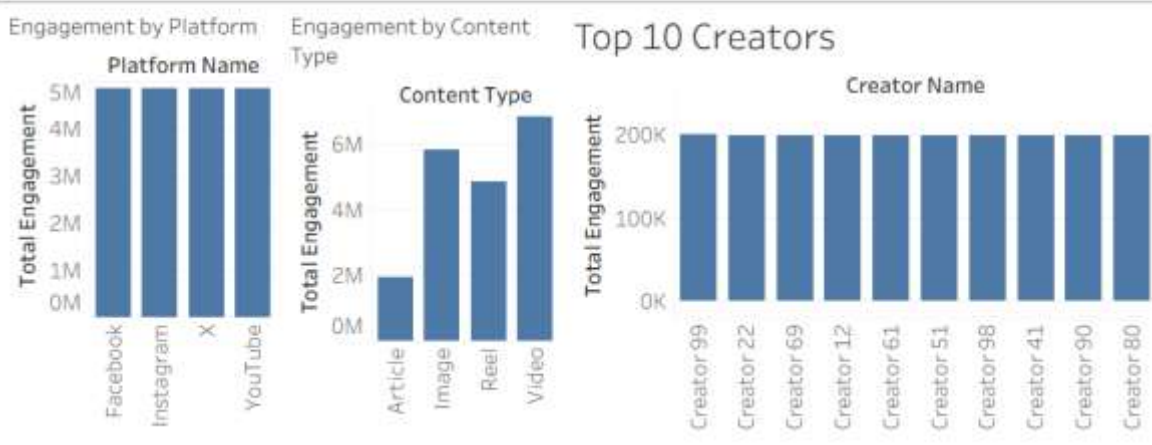
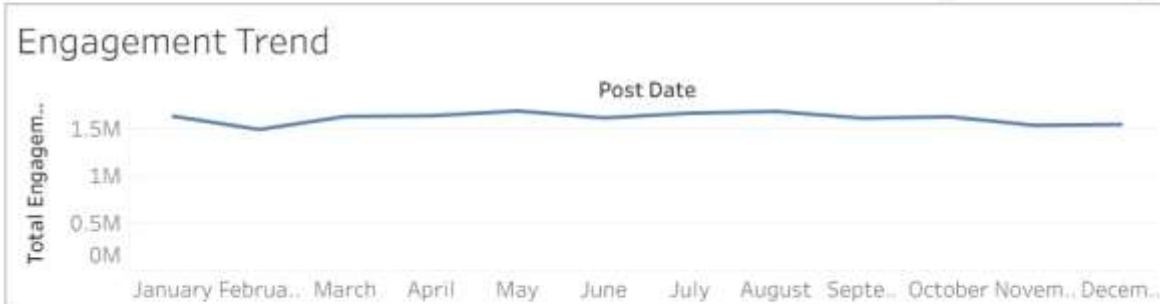
The dashboard contains:

- Total Views KPI
- Total Engagement KPI
- Total Posts KPI
- Engagement Trend
- Engagement by Platform
- Engagement by Content Type
- Top 10 Creators



SOCIAL MEDIA PERFORMANCE DASHBOARD 231401067

Engagement	Views	Posts	Year of Post Date	Platform Name	Content Type
1,93,96,500	14,68,81,500	3,000	☒ (All) ☒ 2024 ☒ 2025 ☒ 2026	☒ Facebook ☒ Instagram ☒ X ☒ YouTube	☒ Article ☒ Image ☒ Reel ☒ Video



Result: